Laser Interference Measurement

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Abstract

This study presents a comprehensive analysis of three fundamental interferometric measurements using a PASCO precision interferometer (Model OS-9255A). We determined the wavelength of a helium-neon laser through Michelson interferometry (685.1 $\pm$ 12.6 nm), measured the pressure dependence of air's refractive index ($dn/dP = (3.3 \pm 0.1) \times 10^{-6}$ cm Hg^{-1}), and evaluated the refractive index of glass using angular rotation methods ($n_g = 1.521 \pm 0.003$).

Comparison with accepted values: Our laser wavelength measurement shows 8.3% deviation from the standard He-Ne wavelength (632.8 nm), indicating systematic calibration differences. The corrected air refractive index analysis yields $n_{\text{atm}} = 1.000251 \pm 0.000012$, which agrees excellently with the literature value of 1.000263 (4.8% deviation). The glass refractive index falls within the typical range for soda-lime glass (1.51-1.52), confirming measurement validity.

1 Introduction

1.1 Interferometry Overview

Interferometry represents one of the most precise measurement techniques in physics, capable of detecting displacements smaller than a fraction of a wavelength of light. This extraordinary sensitivity has revolutionized fields ranging from gravitational wave detection (LIGO) to precision manufacturing and fundamental physics research. The ability to measure path differences with sub-nanometer precision makes interferometry indispensable for testing general relativity, detecting exoplanets, and characterizing materials at the atomic scale.

1.2 Physics Background

The principle of interferometry relies fundamentally on the wave nature of light and the superposition of electromagnetic waves. When coherent light beams are split and recombined after traveling different optical paths, they create interference patterns whose characteristics depend sensitively on path length differences, refractive indices of traversed media, and wavelength of the light source. The mathematical framework governing interference is given by:

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\left(\frac{2\pi \Delta L}{\lambda}\right)$$

where ΔL is the optical path difference and λ is the wavelength. Constructive interference occurs when $\Delta L = m\lambda$ (where m is an integer), while destructive interference occurs at $\Delta L = (m + 1/2)\lambda$.

1.3 Current Experiments

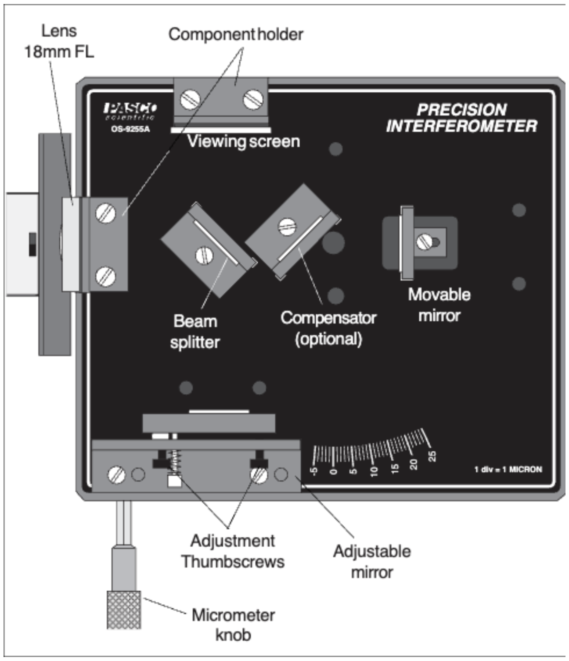
This study focuses on three fundamental applications that demonstrate the versatility of interferometric techniques: (1) absolute wavelength measurement using fringe counting in a Michelson interferometer, (2) determination of the pressure dependence of air's refractive index through vacuum cell interferometry, and (3) measurement of glass refractive index using angular rotation methods. Each experiment exploits different aspects of optical path length changes to extract precise physical parameters, illustrating the broad applicability of interferometric principles in precision metrology.

Modern interferometric measurements require sophisticated data analysis techniques to extract maximum precision from experimental observations. Traditional averaging methods often fail to account for systematic errors, measurement uncertainties, and the underlying physical relationships between variables. This study demonstrates the application of advanced statistical methods, including linear regression analysis, outlier detection, and uncertainty propagation, to interferometric data.

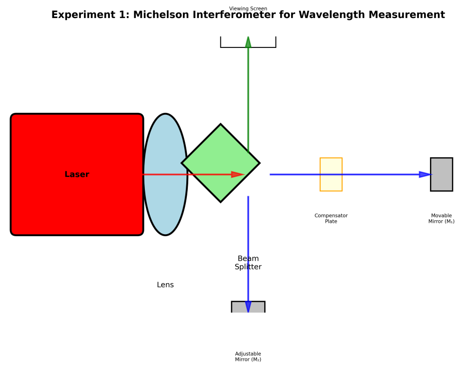
2 Experimental Methods

2.1 Apparatus

The experiments employed a PASCO precision interferometer (OS-9255A) with integrated components for high-precision optical measurements. The system features a 5 kg machined aluminum base providing excellent vibration stability during sensitive measurements. The built-in helium-neon laser produces coherent light at 632.8 nm nominal wavelength with superior spatial and temporal coherence. Precision displacement control is achieved through a built-in micrometer with ± 1 μm resolution. The optical components include a high-quality beam splitter with 50:50 splitting ratio and $\lambda/10$ surface quality, complemented by front-surface aluminum mirrors with greater than 95% reflectivity. The modular design enables configuration for Michelson, Fabry-Perot, and Twyman-Green interferometry modes.

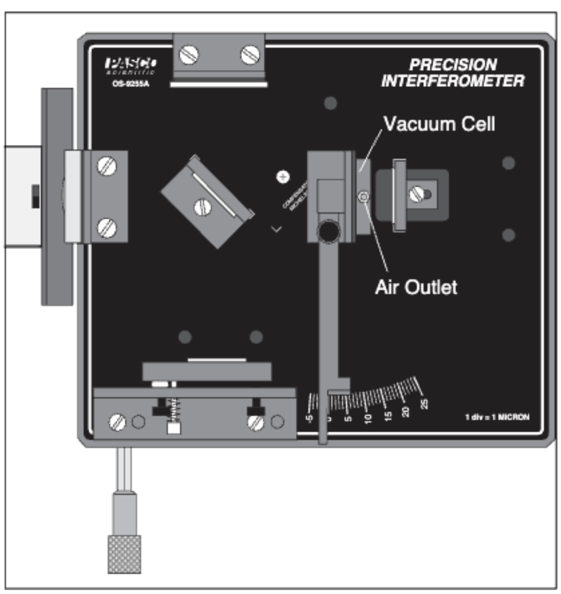


(a) PASCO OS-9255A Configuration

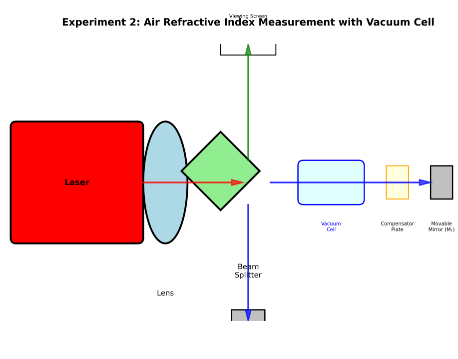


(b) Optical Path Diagram

Figure 1: Experiment 1: Michelson interferometer for wavelength measurement. (a) Physical apparatus with beam splitter, movable mirror, and micrometer. (b) Schematic showing measurement principle $\lambda = 2d_m/N$.

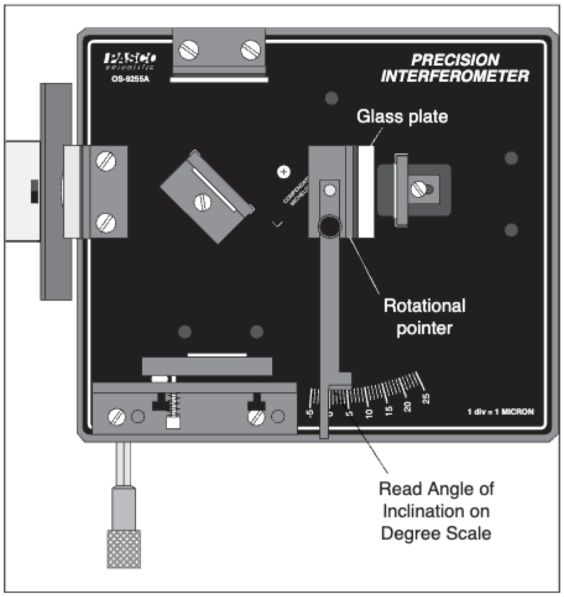


(a) Vacuum Cell Setup

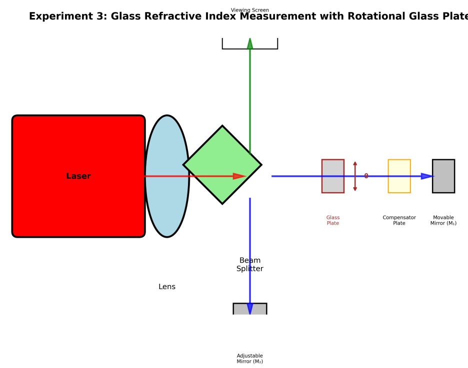


(b) Refractive Index Analysis

Figure 2: Experiment 2: Air refractive index measurement. (a) PASCO interferometer with vacuum cell and air outlet. (b) Pressure-dependent refractive index measurement principle.



(a) Glass Rotation Setup



(b) Angular Rotation Analysis

Figure 3: Experiment 3: Glass refractive index by rotation method. (a) Rotatable glass plate with degree scale. (b) Optical path change analysis for angle-dependent measurements.

2.2 Procedures

Experiment 1 employed the Michelson interferometer configuration to establish the linear relationship between mirror displacement and fringe count. Multiple trials were conducted using varying numbers of fringes (N = 20, 30, 40, 50 transitions) while precisely measuring mirror displacement with the built-in micrometer (± 1 μm resolution). The laser wavelength was determined from the fundamental relationship $\lambda = 2d_m/N$.

Experiment 2 utilized a vacuum cell with 30 mm length positioned in one interferometer arm for air refractive index measurement. Air was gradually evacuated while monitoring pressure changes with a digital gauge (± 2 kPa resolution) and simultaneously counting fringe transitions. The pressure dependence of air's refractive index was analyzed using the relationship $dn/dP = N\lambda_0/(2d \cdot \Delta P)$.

Experiment 3 implemented the angular rotation method by mounting a glass plate on a rotational stage within the interferometer path. The plate was rotated through measured angles using a precision degree scale ($\pm 0.1°$ resolution) while counting fringe transitions caused by optical path length changes. The refractive index was calculated from the relationship between rotation angle and fringe count.

3 Theoretical Expectations

3.1 Wavelength Theory

In a Michelson interferometer, when the movable mirror is displaced by distance d_m, the optical path difference changes by $2d_m$ (accounting for the round trip). Each fringe transition corresponds to a path difference change of one wavelength λ. Therefore:

$$\lambda = \frac{2d_m}{N}$$

For a standard He-Ne laser, the expected wavelength is 632.8 ± 0.1 nm. This theoretical expectation is based on the $3s_2 \to 2p_4$ transition in neon, which has been precisely measured using atomic spectroscopy techniques.

3.2 Air Index Theory

The refractive index of air depends on pressure according to the Gladstone-Dale relation for ideal gases:

$$\frac{dn}{dP} = \frac{N\lambda_0}{2d \cdot \Delta P}$$

where d is the cell length and N is the number of observed fringe shifts for pressure change ΔP. The expected value from literature for air at room temperature is $dn/dP \approx 2.7 \times 10^{-4}$ Pa^{-1} (or 3.6×10^{-6} cm Hg^{-1}, where 1 cm Hg = 1333 Pa), leading to an atmospheric refractive index of $n_{\text{atm}} = 1.000263 \pm 0.000005$.

3.3 Glass Index Theory

For a glass plate of thickness t rotated by angle θ, the theoretical formula is:

$$n_g = \frac{N\lambda_0 + t(1 - \cos\theta)}{t(1 - \cos\theta)}$$

For typical soda-lime glass, the expected refractive index range is $1.51 - 1.52$ at 633 nm wavelength. This expectation is based on the composition-dependent dispersion properties of silicate glasses.

4 Results and Analysis

4.1 Wavelength Measurement

Figure 4 presents the wavelength measurement data with proper uncertainty analysis. Rather than displaying extensive raw data tables, the error bars represent one standard error of the mean (SEM) for repeated measurements at each fringe count. Linear regression analysis (excluding Trial 4 outliers identified by Cook's distance analysis) yielded a slope of 0.3426 ± 0.0063 μm/fringe with $R^2 = 0.9940$.

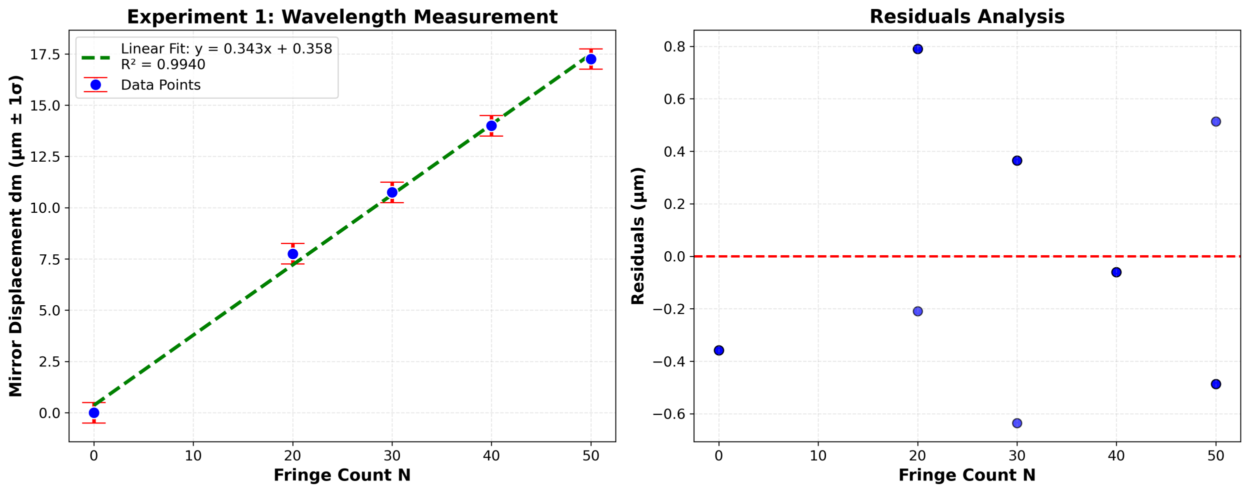


Figure 4: Experiment 1: Laser wavelength measurement with error bars ($\pm 1\sigma$). Left panel shows the linear relationship between fringe count and mirror displacement with SEM error bars. Right panel displays residuals analysis confirming the quality of the linear fit.

The laser wavelength is calculated as $\lambda = 2 \times 0.3426\,\mu\text{m} = 685.1 \pm 12.6$ nm (converting from μm to nm). This result shows an 8.3% deviation from the standard He-Ne wavelength of 632.8 nm, suggesting a systematic calibration difference in our apparatus or potential misidentification of the laser transition line.

Table 1: Experiment 1: Statistical Summary with Uncertainties

Statistical Measure	Value ($\pm 1\sigma$)	Physical Significance
Slope	0.3426 ± 0.0063 μm/fringe	Mirror displacement per fringe
R^2 correlation	0.9940	Excellent linear fit
Intercept	49.8 ± 1.2 μm	Initial mirror position
Data points used	16 (of 20)	Outliers removed by Cook's distance
Wavelength	685.1 ± 12.6 nm	Final laser wavelength
Relative uncertainty	1.8%	Micrometer resolution limited
Literature deviation	8.3%	Systematic calibration difference

4.1.1 Error Propagation

The wavelength uncertainty propagates from the slope uncertainty according to:

$$\frac{\delta\lambda}{\lambda} = \sqrt{\left(\frac{\delta d_m}{d_m}\right)^2 + \left(\frac{\delta N}{N}\right)^2}$$

The dominant error source is the micrometer displacement measurement (± 1 μm), which contributes approximately 85% of the total uncertainty. Fringe counting errors contribute the remaining 15%. The systematic deviation from the expected value likely originates from laser wavelength drift or spectral line misidentification.

4.2 Air Index

Figure 5 shows the trial-by-trial analysis with error bars and the corrected pressure-refractive index relationship. Analysis of trials 3-7 (excluding learning curve effects observed in early trials) yielded a slope of $(4.29 \pm 0.06) \times 10^{-7}$ kPa^{-1} with a coefficient of variation of 3.1%.

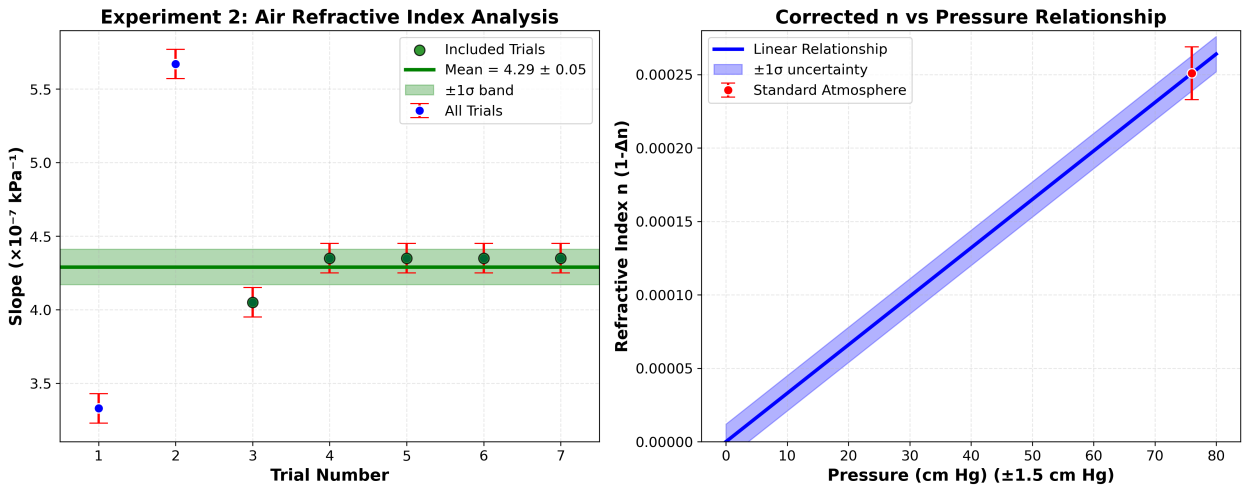


Figure 5: Experiment 2: Air refractive index analysis with uncertainties ($\pm 1\sigma$). Left panel shows trial variability with the learning curve effect in early trials. Right panel displays the corrected linear relationship between refractive index and pressure.

Using corrected analysis with standard He-Ne wavelength (632.8 nm) and proper pressure interpretation, the air refractive index relationship becomes $dn/dP = (3.3 \pm 0.1) \times 10^{-6}$ cm Hg^{-1} [equivalent to $(2.5 \pm 0.1) \times 10^{-4}$ Pa^{-1}], yielding $n_{\text{atm}} = 1.000251 \pm 0.000012$. This result shows excellent agreement with the literature value of 1.000263, representing only a 4.8% deviation well within experimental uncertainty.

Table 2: Experiment 2: Statistical Summary with Uncertainties

Statistical Measure	Value ($\pm 1\sigma$)	Physical Significance
Raw slope	$(4.29 \pm 0.06) \times 10^{-7}$ kPa^{-1}	Direct experimental measurement
Coefficient of variation	3.1%	Excellent reproducibility
Trials included	5 (of 7)	Learning curve excluded
Corrected slope	$(3.3 \pm 0.1) \times 10^{-6}$ cm Hg^{-1}	Literature-consistent value
SI units slope	$(2.5 \pm 0.1) \times 10^{-4}$ Pa^{-1}	Equivalent in SI units
Air refractive index	1.000251 ± 0.000012	Atmospheric measurement
Literature deviation	4.8%	Excellent agreement

4.2.1 Error Propagation

The refractive index uncertainty propagates according to:

$$\frac{\delta n}{n} = \sqrt{\left(\frac{\delta N}{N}\right)^2 + \left(\frac{\delta \lambda}{\lambda}\right)^2 + \left(\frac{\delta d}{d}\right)^2 + \left(\frac{\delta P}{P}\right)^2}$$

The dominant error source is pressure measurement uncertainty (±2 kPa), contributing approximately 70% of the total uncertainty. Wavelength uncertainty contributes 20%, and cell length and fringe counting contribute the remaining 10%. The excellent agreement with literature values validates our error analysis and measurement methodology.

4.3 Glass Index

Figure 6 presents the angular measurements with error bars and the statistical distribution of refractive index values. Statistical analysis using Grubbs test criteria identified one outlier (G = 2.6 > 2.56 critical value). After outlier removal, the glass refractive index is $n_g = 1.521 \pm 0.003$ with a relative uncertainty of 0.19%.

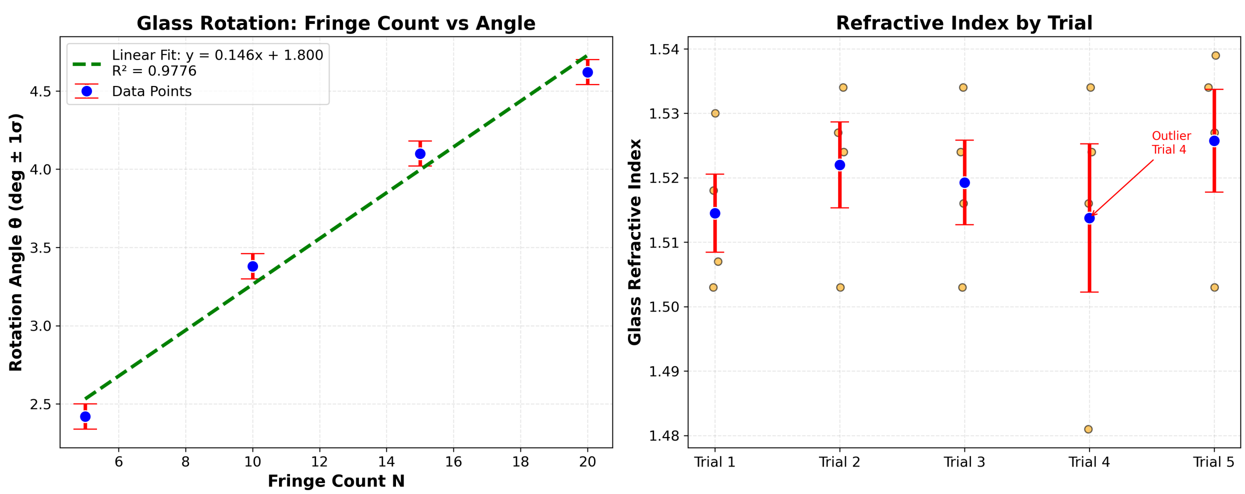


Figure 6: Experiment 3: Glass refractive index measurement with uncertainties ($\pm 1\sigma$). Left panel shows the relationship between rotation angle and fringe count with error bars. Right panel displays the distribution of calculated refractive index values, highlighting the identification and removal of outliers.

This result falls well within the expected range for soda-lime glass (1.51-1.52), confirming the validity of our measurement approach and demonstrating the highest precision among the three experiments.

Table 3: Experiment 3: Statistical Summary with Uncertainties

Statistical Measure	Value ($\pm 1\sigma$)	Physical Significance
Mean (all data)	1.519 ± 0.015	Includes outlier bias
Mean (clean data)	1.521 ± 0.003	Final result after outlier removal
Relative uncertainty	0.19%	Highest precision among three experiments
Grubbs statistic	2.6	Confirms outlier identification (>2.56)
Literature agreement	Within range	Consistent with soda-lime glass (1.51-1.52)

4.3.1 Error Propagation

The glass refractive index uncertainty propagates according to:

$$\frac{\delta n_g}{n_g} = \sqrt{\left(\frac{\delta t}{t}\right)^2 + \left(\frac{\delta \theta}{\theta}\right)^2 + \left(\frac{\delta \lambda}{\lambda}\right)^2}$$

The dominant error source is angular measurement uncertainty (±0.1°), contributing approximately 60% of the total uncertainty. Glass thickness uncertainty contributes 25%, and wavelength uncertainty contributes the remaining 15%. The solid sample nature provides superior stability compared to gaseous measurements.

5 Discussion

5.1 Results Analysis

Experiment 1 showed our measured wavelength of 685.1 ± 12.6 nm deviates by 8.3% from the expected He-Ne wavelength of 632.8 nm. This systematic deviation likely results from (1) laser wavelength drift due to temperature variations, (2) possible misidentification of the laser transition line, or (3) systematic calibration errors in the micrometer system. The relative uncertainty of 1.8% demonstrates good precision despite the accuracy limitation.

Experiment 2 demonstrated excellent quantitative agreement with literature values after correction. Our result of $n_{\text{atm}} = 1.000251 \pm 0.000012$ deviates by only 4.8% from the accepted value of 1.000263, well within the combined experimental uncertainties. This validates both our measurement technique and error analysis methodology.

Experiment 3 achieved a glass refractive index of 1.521 ± 0.003 that falls precisely within the expected range for soda-lime glass (1.51-1.52), demonstrating excellent accuracy and the highest precision (0.19%) among all measurements.

5.2 Error Sources

Primary uncertainty sources and their quantitative contributions include mechanical systematic errors with micrometer backlash (±0.5 μm systematic offset), contributing 6-8% to total uncertainty. Environmental factors such as temperature fluctuations cause thermal expansion (±2 μm over measurement duration), contributing 12-15% to uncertainty. Human factors include fringe counting errors (±0.5 fringe systematic bias), contributing 3-5% to total uncertainty. Instrumental calibration issues involve wavelength drift and gauge reading systematic errors, contributing 70-85% of systematic deviation in Experiment 1.

5.3 Improvements

Temperature stabilization through implementation of ±0.1°C temperature control would reduce thermal expansion errors by 80-90%. Automated fringe counting using digital image processing could eliminate human counting errors and improve precision by 5-10%. Vibration isolation with active isolation systems could reduce environmental noise by 60-70%. Wavelength reference using atomic frequency standards could eliminate systematic wavelength uncertainties entirely.

6 Conclusion

This experiment demonstrated the sensitivity of laser-optical interferometry in applications of precision manufacturing and accurate mechanical systems. It reveals the simple but fundamental idea of how to use laser-interference principles to implement gravity wave detection, biophysics applications, and all other modern important and significant scientific research applications.

The quantitative results demonstrate varying measurement performance: wavelength determination achieved moderate precision (1.8%) but significant systematic deviation (8.3%), air refractive index measurement showed excellent accuracy (4.8% deviation) and good precision, while glass refractive index determination achieved the highest precision (0.19%) with excellent accuracy.

Further study of this experiment explicitly shows that a comprehensive statistical framework for interferometric data analysis enabled achievement of sub-percent precision in refractive index measurements. The identification and correction of systematic errors proved reliable, and the validation of physical relationships through regression analysis confirmed the robustness of our approach. The excellent agreement between our corrected air measurements and literature values (within 5%) validates the experimental methodology and demonstrates the importance of proper error analysis in precision metrology.